



Liver, Biliary System, Pancreas & Spleen — MCQ Practice

Anatomy GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time $\approx$ 25 minutes.

Q Questions

1. Roughly what proportion of the liver's blood supply comes from the portal vein?

EASY

- A. 30%
- B. 70%
- C. 50%
- D. 10%
- E. 90%

2. How much bile does the gallbladder store?

EASY

- A. 5-10 mL
- B. 100-150 mL
- C. 500 mL
- D. 30-50 mL
- E. 1 litre

3. Which part of the pancreas projects as the uncinate process?

EASY

- A. Lower part of the head
- B. Neck
- C. Body
- D. Tail
- E. Upper part of the head

4. Which line divides the liver into two functional (physiological) lobes?

EASY

- A. Falciform ligament
- B. Fissure for ligamentum venosum
- C. Cantlie's line
- D. Coronary ligament
- E. Porta hepatis

5. During liver surgery the surgeon examines the hilum between the caudate and quadrate lobes. Which structure is NOT among its contents?

MEDIUM

- A. Right and left hepatic ducts
- B. Branches of the hepatic artery
- C. Branches of the portal vein
- D. Hepatic lymph nodes
- E. Hepatic veins

6. A pathologist traces lymph from the bare area of the liver. Where does it drain?

MEDIUM

- A. Through the diaphragm to the posterior mediastinal nodes
- B. To the superficial inguinal nodes
- C. To the superior mesenteric nodes
- D. To the axillary nodes
- E. To the pancreaticosplenic nodes

7. A 45-year-old woman has acute cholecystitis. Besides right upper quadrant pain, where is pain classically referred, according to the lecture?

MEDIUM

- A. Umbilicus
- B. Left iliac fossa
- C. Shoulder
- D. Groin
- E. Back of the thigh

8. During laparoscopic cholecystectomy the surgeon identifies a triangle bounded by the cystic duct, cystic artery and common hepatic duct. Which is it?

MEDIUM

- A. Hepatocystic triangle
- B. Calot's triangle
- C. Lumbar triangle
- D. Hesselbach's triangle
- E. Triangle of Koch

9. Why is the gallbladder prone to gangrene when its artery is occluded?

MEDIUM

- A. It has no venous drainage
- B. It is retroperitoneal
- C. Its wall has no muscle
- D. It is supplied by a single (cystic) artery
- E. It lies above the transpyloric plane

10. Where is the portal vein formed, and by which veins?

MEDIUM

- A. Behind D2 by the IMV and splenic vein
- B. In the porta hepatis by the right and left gastric veins
- C. Behind the head of the pancreas by the SMV and IMV
- D. In front of the IVC by the hepatic veins
- E. Behind the neck of the pancreas by the SMV and splenic vein

11. A patient with portal hypertension has dilated veins radiating from the umbilicus. Which portal vein forms this anastomosis?

MEDIUM

- A. Paraumbilical veins
- B. Left gastric vein
- C. Superior rectal vein
- D. Splenic vein
- E. Right gastric vein

12. A tumour of the head of the pancreas compresses a structure that runs behind it, causing obstructive jaundice. Which structure?

MEDIUM

- A. Splenic artery
- B. Left renal vein
- C. Common bile duct
- D. Superior mesenteric artery
- E. Left crus of the diaphragm

13. Which structure runs along the superior border of the body of the pancreas?

MEDIUM

- A. Superior mesenteric vein
- B. Inferior mesenteric vein
- C. Left renal vein
- D. Splenic artery
- E. Gastroduodenal artery

14. During splenectomy the surgeon ligates the vessels in the ligament through which the tail of the pancreas reaches the splenic hilum. Which ligament is it?

MEDIUM

- A. Gastrosplenic ligament
- B. Splenorenal (lienorenal) ligament
- C. Phrenicocolic ligament
- D. Falciform ligament
- E. Coronary ligament

15. A CT shows a tumour of the uncinate process. Between which two structures does the uncinate process lie?

HARD

- A. IVC behind and CBD in front
- B. Splenic vein behind and stomach in front
- C. Left kidney behind and transverse colon in front
- D. Portal vein behind and gastroduodenal artery in front
- E. Abdominal aorta behind and superior mesenteric vessels in front

16. An injury damages the accessory pancreatic duct. Which part of the pancreas will have impaired drainage, and where does this duct open?

HARD

- A. Uncinate process and lower head; second part of duodenum above the ampulla of Vater
- B. Tail and body; at the ampulla of Vater
- C. Upper head only; into the common bile duct
- D. Neck; first part of duodenum
- E. Whole gland; third part of duodenum

17. A posterior perforation of the stomach leaks into the space directly in front of the body of the pancreas. Which space separates the stomach from the anterior surface of the pancreatic body?

HARD

- A. Greater sac
- B. Right subphrenic space
- C. Hepatorenal pouch
- D. Lesser sac
- E. Right paracolic gutter

18. A cancer of the upper head of the pancreas spreads by lymphatics. Which node group is first involved?

HARD

- A. Superior mesenteric nodes
- B. Pancreaticosplenic nodes
- C. Celiac nodes
- D. Inferior mesenteric nodes
- E. Superficial inguinal nodes

19. A man with long-standing portal hypertension has massive haematemesis, rectal bleeding and caput medusae. Which pairing of portal and caval veins is correct?

HARD

- A. Superior rectal ↔ azygos
- B. Left gastric ↔ inferior rectal
- C. Paraumbilical ↔ azygos
- D. Splenic ↔ renal
- E. Left gastric ↔ azygos; superior rectal ↔ middle and inferior rectal; paraumbilical ↔ abdominal wall veins

20. During splenic surgery, which statement about the splenic vessels is correct?

HARD

- A.** The splenic vein is more tortuous than the artery
- B.** The splenic vein usually lies behind the artery and receives the inferior mesenteric vein
- C.** The splenic artery arises from the SMA
- D.** The splenic vein drains into the IVC
- E.** The splenic artery enters the hilum through the gastrocolic ligament

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	The portal vein supplies about 70% of the liver's blood; the hepatic artery about 30%.
2	D	EASY	The gallbladder lies under the liver and stores 30–50 mL of bile.
3	A	EASY	The lower part of the head of the pancreas forms a projection called the uncinate process.
4	C	EASY	Cantlie's line (principal plane) divides the liver into two functional lobes; the falciform ligament divides the anatomical lobes.
5	E	MEDIUM	The porta hepatis contains hepatic ducts, hepatic artery and portal vein branches, nerves and lymph nodes. Hepatic veins leave the liver posteriorly to the IVC.
6	A	MEDIUM	Lymph from the porta hepatis goes to the celiac nodes; lymph from the bare area passes through the diaphragm to the posterior mediastinal nodes.
7	C	MEDIUM	The slides list acute cholecystitis with referred pain to the shoulder.
8	B	MEDIUM	Calot's triangle is bounded by the cystic duct, cystic artery and common hepatic duct. The hepatocystic triangle has the liver edge as its upper boundary and contains the cystic artery.
9	D	MEDIUM	The slides list gangrene of the gallbladder as a risk because of its single arterial supply, the cystic artery.
10	E	MEDIUM	The portal vein is formed behind the neck of the pancreas by the union of the superior mesenteric and splenic veins.
11	A	MEDIUM	At the umbilicus the paraumbilical veins (portal) anastomose with the superficial veins of the anterior abdominal wall (caval).
12	C	MEDIUM	Behind the head of the pancreas lie the IVC, renal veins and common bile duct; compression of the CBD causes obstructive jaundice.
13	D	MEDIUM	The superior border of the body is related to the splenic artery; the anterior border gives attachment to the transverse mesocolon.
14	B	MEDIUM	The tail of the pancreas reaches the splenic hilum via the lienorenal ligament, and the splenic artery enters the hilum between its layers.
15	E	HARD	The uncinate process lies between the abdominal aorta and the superior mesenteric vessels.
16	A	HARD	The accessory duct drains the uncinate process and lower head and opens into D2 above the ampulla of Vater; the main duct drains the upper head, body and tail.
17	D	HARD	The anterior surface of the body of the pancreas is related to the stomach, separated from it by the lesser sac.
18	C	HARD	Upper head → celiac nodes; lower head → superior mesenteric nodes; rest of the pancreas → pancreaticosplenic nodes.
19	E	HARD	Lower esophagus: left gastric ↔ azygos. Upper anal canal: superior rectal ↔ middle and inferior rectal. Umbilicus: paraumbilical ↔ superficial veins of the anterior abdominal wall.
20	B	HARD	The splenic vein lies usually posterior to the artery, is less tortuous, receives the IMV and joins the SMV to form the portal vein. The very tortuous artery enters through the splenorenal ligament.